COMPUTER ENGINEERING

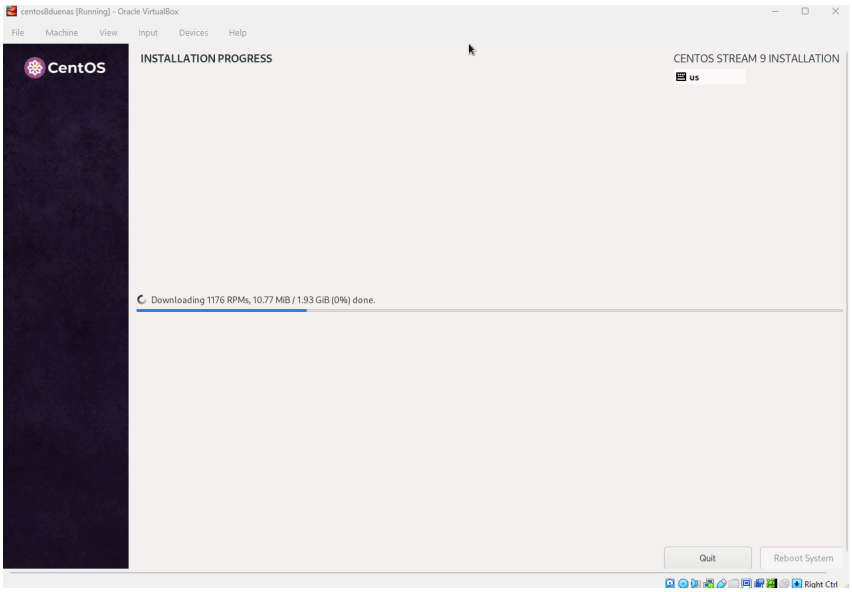
QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	01-08-2026	**DATE SUBMITTED:**	01-08-2026
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ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image



Task 2: Install the SSH server package openssh

. Install the ssh server package openssh by using the dnf command:

$ dnf install openssh-server

```
[duenas@localhost ~]$ sudo dnf install openssh-server
Last metadata expiration check: 0:01:07 ago on Sat 01 Aug 2026 10:03:06 AM PST.
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

2. Start the sshd daemon and set to start after reboot:

$ systemctl start sshd

$ systemctl enable sshd

```
[duenas@localhost ~]$ systemctl start sshd
[duenas@localhost ~]$ systemctl enable sshd
[duenas@localhost ~]$
```

3. Confirm that the sshd daemon is up and running:
$ systemctl status sshd

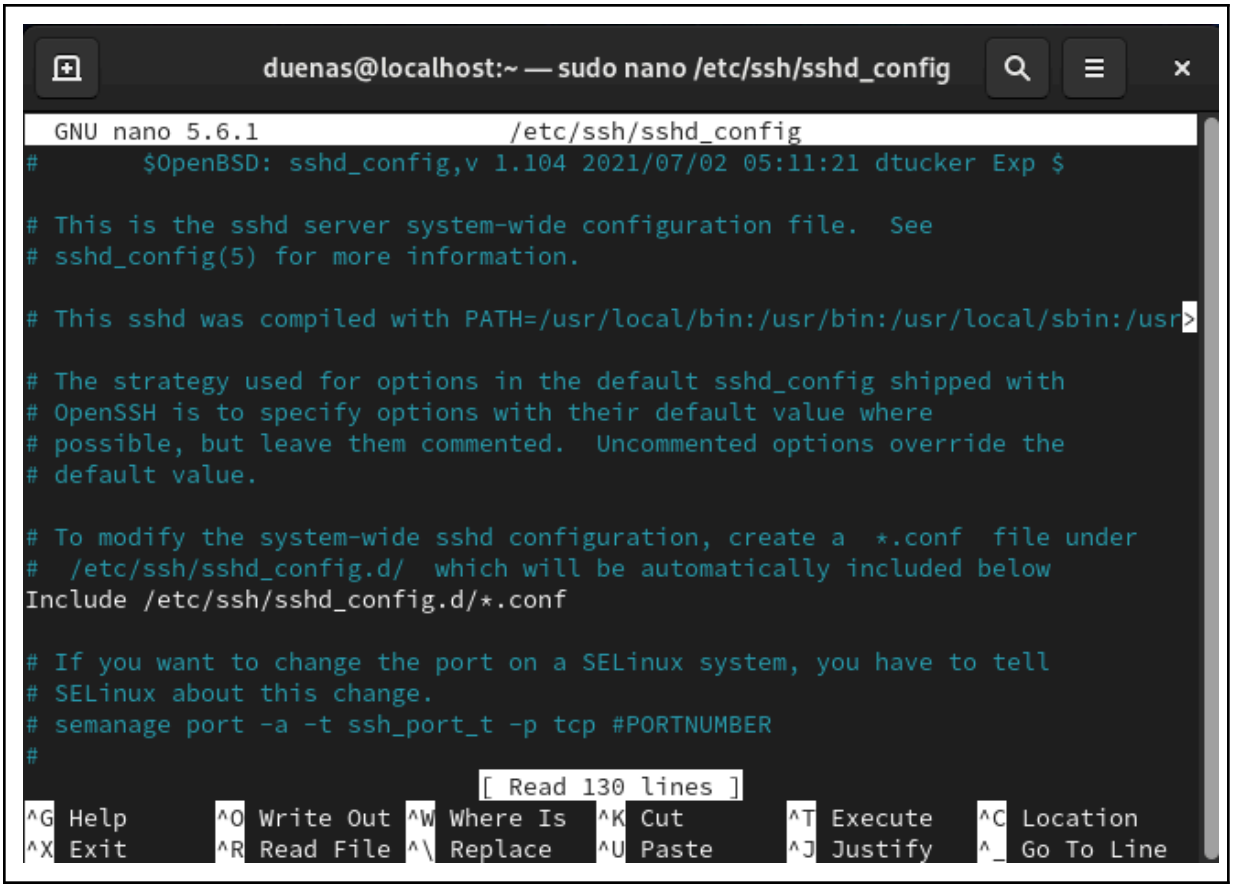
```
[duenas@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Sat 2026-08-01 10:02:41 PST; 2min 40s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 975 (sshd)
      Tasks: 1 (limit: 34056)
     Memory: 2.1M (peak: 2.4M)
        CPU: 13ms
     CGroup: /system.slice/sshd.service
             └─975 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 01 10:02:41 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Aug 01 10:02:41 localhost.localdomain sshd[975]: Server listening on 0.0.0.0 po>
Aug 01 10:02:41 localhost.localdomain sshd[975]: Server listening on :: port 22.
Aug 01 10:02:41 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

4. Open the SSH port 22 to allow incoming traffic:
$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload

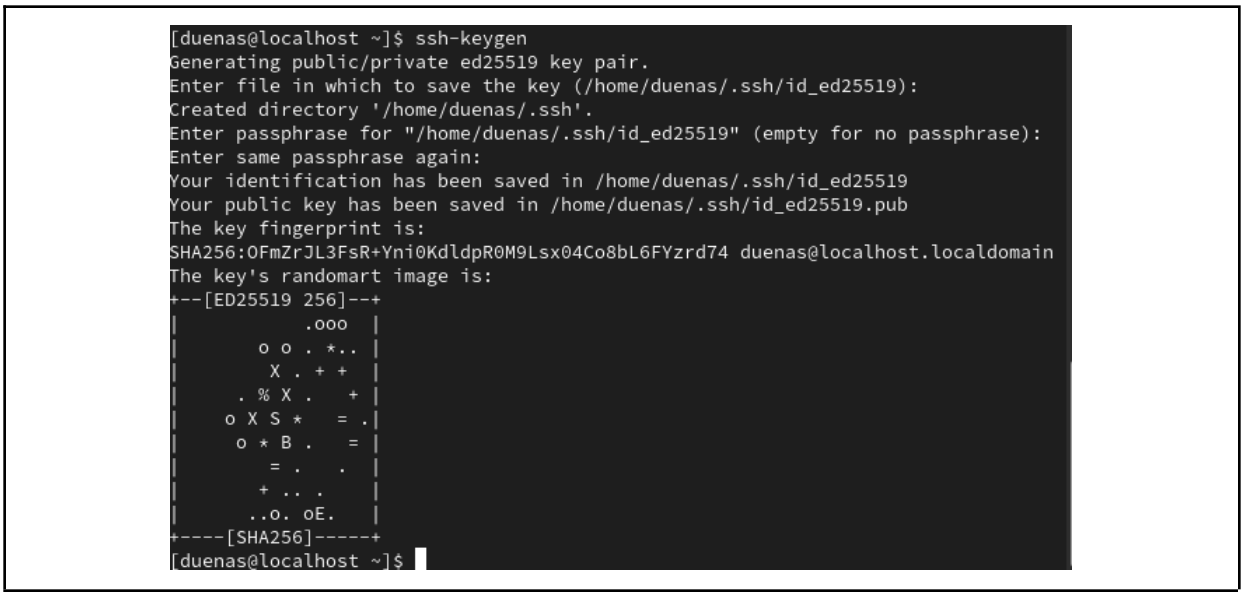
```
[duenas@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[duenas@localhost ~]$ firewall-cmd --reload
success
[duenas@localhost ~]$
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:
$ systemctl reload sshd



Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.



2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
[duenas@localhost ~]$ ssh-copy-id duenas@10.0.2.15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist
on the remote system.
                (if you think this is a mistake, you may want to use -f option)

[duenas@localhost ~]$
```

3. On CentOS, verify that you have the authorized_keys.

```
[duenas@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMq53B7/pKcHDeAXoWVLoRDJBv2vEfJbp1LmNhwnzTCT
 duenas@localhost.localdomain
[duenas@localhost ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

```
[duenas@localhost ~]$ ssh duenas@10.0.2.15
Activate the web console with: systemctl enable --now cockpit.socket

Last failed login: Sat Aug  1 10:19:03 PST 2026 from 10.0.2.15 on ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Sat Aug  1 10:15:10 2026
```

2. Show evidence that you are connected

```
Last login: Sat Aug  1 10:15:10 2026
[duenas@localhost ~]$ whoami && hostname && cat /etc/redhat-release
duenas
localhost.localdomain
CentOS Stream release 9
[duenas@localhost ~]$ who
duenas   tty2         2026-08-01 10:15 (local)
duenas   pts/1        2026-08-01 10:21 (10.0.2.15)
[duenas@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- check whether you actually need commercial vendor support, versus some community driven model that kind of figures itself out over time. Also consider your preferences, like software stability, the target deployment environment, and how you feel about the package management ecosystem.

2. What are the main differences between Debian and Red Hat Linux distributions?
   - Debian comes with .deb packages, handled via apt, and it defaults to AppArmor security in a community led setup. Red Hat instead uses .rpm packages with dnf, defaults to SELinux security, and usually comes with predictable enterprise grade commercial support.

Conclusion

Today we resolved that SSH identity error by generating a new key pair using ssh-keygen, which enabled passwordless public key authentication for your CentOS box. After that we compared Debian and Red Hat Linux, we highlighted their big distinctions in package management, security frameworks and enterprise support.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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